Rockchip Linux6.12 SDK Release Note

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Preface

Overview

The document presents Rockchip Linux6.12 SDK release notes, aiming to help engineers get started with Linux6.12 universal SDK development and related debugging faster.

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Chipset and System Support

Chipset	Buildroot	Debian	Yocto
RK3572	Y	Y	Y
RK3576	Y	Y	Y
RK3588	Y	Y	Y

Revision History

Date	Version	Author	Revision History
2026-03-20	V0.0.1	Weiye Chen	Alpha version
2026-03-30	V0.0.2	Weiye Chen	Remove Hardware Development Guide
2026-04-20	V0.1.0	Weiye Chen	Beta version
2026-06-20	V1.0.0	Weiye Chen	1.Release version 2.Add RK3576/RK3588 support

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1. Overview

This SDK supports three systems based on Buildroot 2026.02, Debian 13, and Yocto 6.0, with the kernel based on Kernel 6.12 and the bootloader based on U-Boot v5. It is suitable for Rockchip EVB development boards and all Linux products developed based on these boards.

The development kit is applicable to, but not limited to, AIoT solution-related application scenarios and products such as industrial motherboards, robots, NVRs, drones, and conference audio/video systems. It provides flexible data path combination interfaces to meet customers' customized needs for free combination. For specific function debugging and interface descriptions, please refer to the documents under the project's docs/ directory.

2. Main Functions

Function	Module Name
System	Buildroot, Debian, Yocto
Partition Table	uboot, misc, boot, recovery, rootfs, oem, userdata
File System Types	EXT2/3/4, VFAT, NTFS, UBIFS, SquashFS
Upgrade Recovery	OTA, AB, Recovery
Secure Boot	Secure Boot
Stress Test Tool	ROCKCHIP_TEST
Data Communication	Wi-Fi, Ethernet card, USB, SD card, SATA, PCIe
Applications	Multimedia Playback, Camera Preview, Desktop UI, Browser

3. How to Get the SDK

The SDK is distributed via Rockchip's official code repository and can only be downloaded with **authorized access**. For offline acquisition, you may request a **local source code package** (contact Rockchip's technical support team for assistance). For details on the compilation and development environment, refer to Section 4 Software Development Guide.

3.1 Download Authorization

If you have completed account creation and need to generate your own HTTPS password before downloading the code, download the repository `https://gerrit.rock-chips.com:8443/repo-release/tools/script` and run `script/Generate-Credential.x` as shown below.

```
mkdir ~/rk_gerrit_password
cd ~/rk_gerrit_password
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/script
cd ./script
./Generate-Credential.x
```

The log of running the authorization script is as follows:

```
Adding configuration for Gerrit server to /home/Test123/rk_gerrit_password
********************************************************************************************************
********************************************************************************************************
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Do you agree to the above terms? (yes/no) yes
Please input your username (make sure to use the username exactly the same as the one in RK email!!): Test123
Please input your email (make sure to use the company email address!!): Test123@rock-chips.com
Please input your SSH private key filename (e.g. id_rsa):
Your Credential check has passed
New git credential will be added into your history!!
You are All Set!!
```

3.2 Get Source Code from Rockchip Code Server

To get the Linux6.12 SDK software package, customers need an account to access the source code repository provided by Rockchip. To synchronize code, provide an SSH public key for server authentication and authorization when applying for the SDK from Rockchip technical support. For Rockchip server SSH public key authorization, refer to Section 5 SSH Public Key Operation Introduction.

Linux6.12 SDK download command is as follows:

Download address:

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo \
-u https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b \
linux6.12 -m rockchip_linux6.12_release.xml
```

Server mirror download:

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo \
-u https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b \
linux6.12 -m rockchip_linux6.12_release.xml --mirror
```

repo is a script written by Google in Python that calls git, mainly used to download and manage software repositories of projects. The download address is as follows:

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

3.3 Get Source Code from Local Compression Package

For quick access to SDK source code, Rockchip technical support usually provides the corresponding SDK initial compression package. In this way, developers can obtain the SDK source code by decompressing the initial compression package, which, after synchronization, is the same as the code downloaded by repo. Take ROCKCHIP_LINUX6.12_SDK_RELEASE_V1.0.0_20260620.tgz as an example. After getting an initialization package, you can obtain the source code by running the following command:

```
mkdir rockchip_linux6.12
tar xvf ROCKCHIP_LINUX6.12_SDK_RELEASE_V1.0.0_20260620.tgz -C rockchip_linux6.12
cd rockchip_linux6.12
.repo/repo/repo sync -l
.repo/repo/repo sync -c
```

Developers can update via `.repo/repo/repo sync -c` command according to update instructions that are regularly released by FAE window.

If you encounter issues downloading the repository, you can use the `--force-sync` parameter to force an update, as in `.repo/repo/repo sync -c --force-sync`. Before doing so, please ensure that any local modifications have been backed up.

After updating the SDK code, a clean operation is required. For example: `./build.sh cleanall`

> Note:
> The software release version can be viewed through engineering XML, as follows:

```
.repo/manifests$ realpath rockchip_linux6.12_release.xml
```

For example, if the printed version is v1.0.0 and the update time is 20260620, the corresponding path is as follows:

```
<SDK>/.repo/manifests/release/rockchip_linux6.12_release_v1.0.0_20260620.xml
```

4. Software Development Guide

For software development, please refer to the quick start documents in the project directory:

```
<SDK>/docs/en/Release/Rockchip_Quick_Start_Linux6.12_EN.pdf
```

5. SSH Public Key Operation Introduction

Please follow “Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf” to generate an SSH public key, and send it via email to sw.fae@rock-chips.com to obtain the SDK code.
This SDK application guide will be released to customers during the permission application process.

5.1 Key Authority Management

The server monitors download times and IP information of each key in real time. If an abnormality is found, download permission of the corresponding key will be disabled.

Keep the private key file properly. Do not grant second authorization to third parties.

5.2 Reference Documents

For more details, please refer to
“/docs/en/Others/Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf”